

Development of Respiratory System

➤ Development of Respiratory System

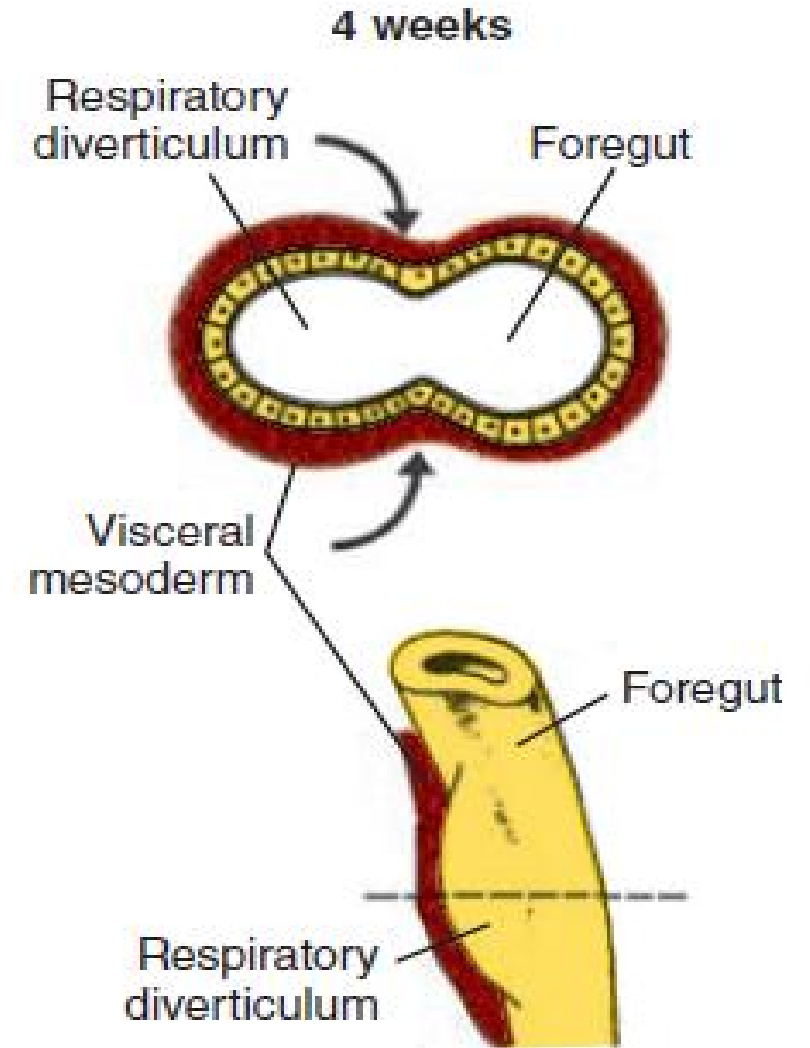
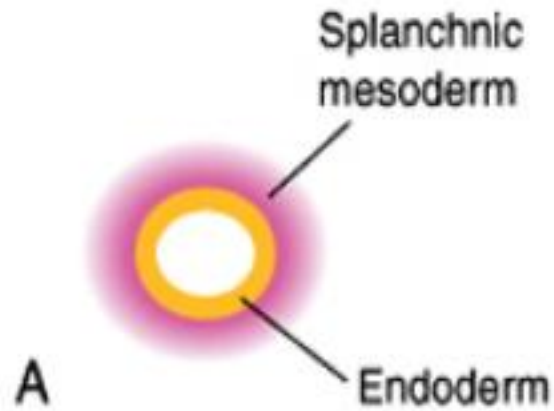
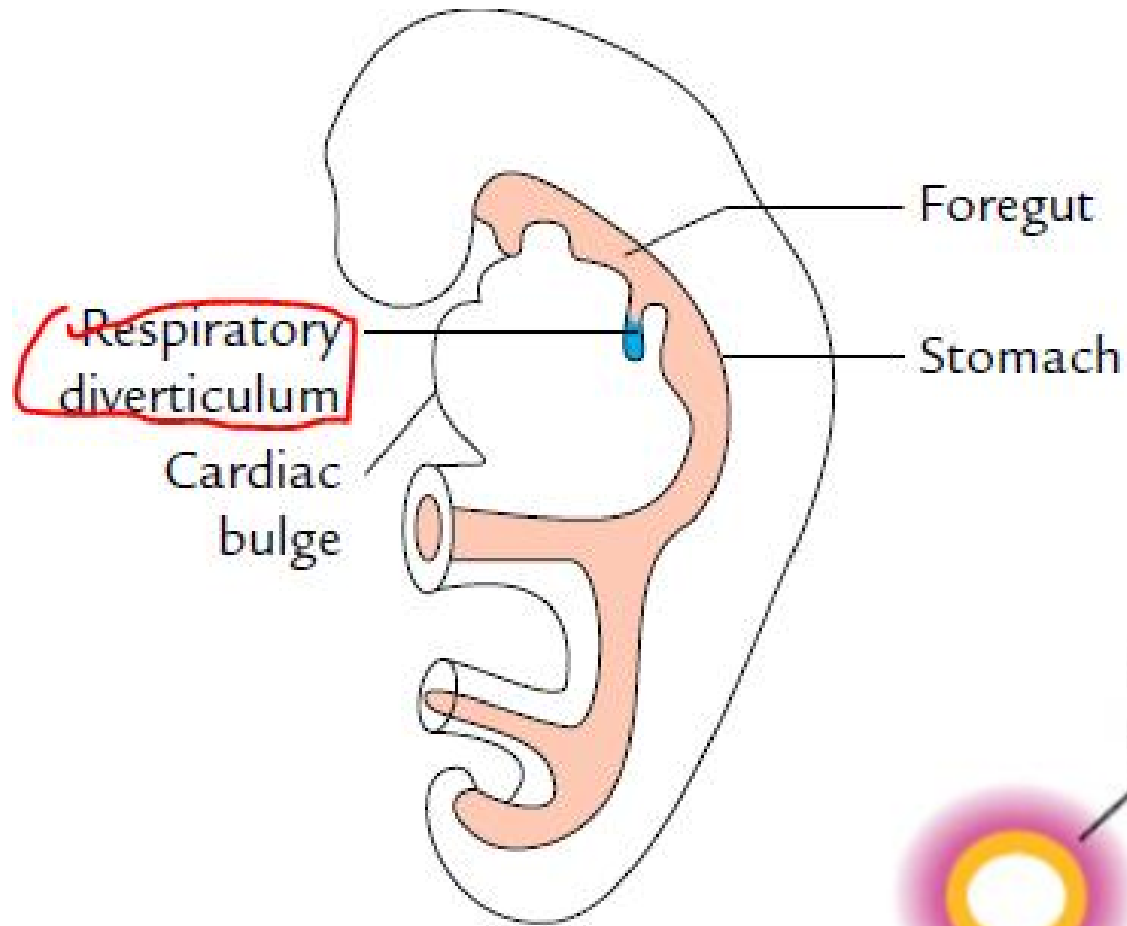
- ✓ Respiratory bud,
- ✓ Lung buds
- ✓ Lung maturation periods (pseudoglandular, canalicular, terminal saccular and alveolar periods)
- ✓ Congenital anomalies of respiratory system
 - Tracheoesophageal fistulae
 - Respiratory distress syndrome

Introduction to Development of Respiratory System

➤ *Structures of respiratory are derived from different origins:*

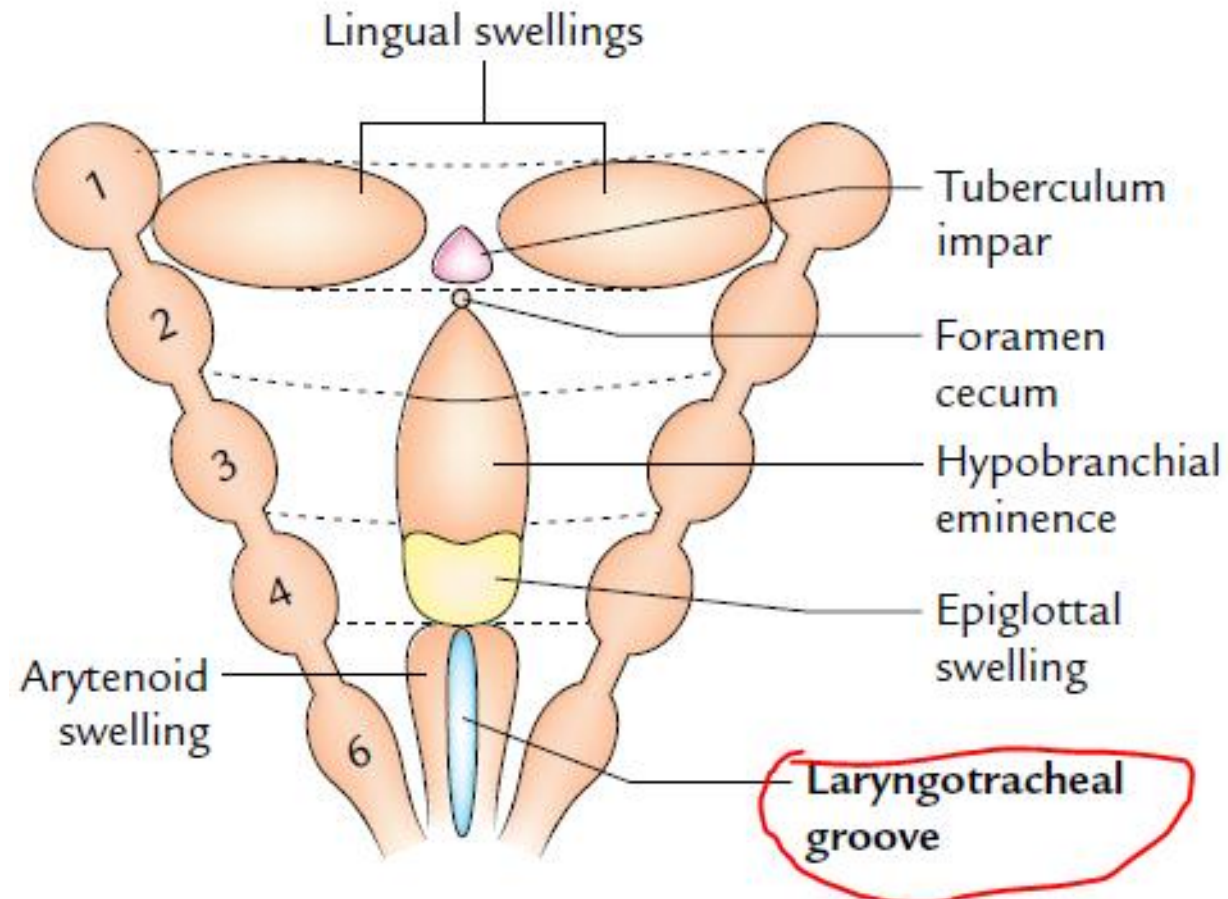
- ❖ Most of its *structures* developed from a **median diverticulum of foregut called respiratory diverticulum.**
- ✓ **From endoderm germ layer:** epithelial lining of trachea, larynx, bronchi, alveoli
- ✓ **From splanchnic mesoderm:** cartilage, muscle, and connective tissue of tract and visceral pleura.

Introduction to Development of Respiratory System,,,



Introduction to Development of Respiratory System,,

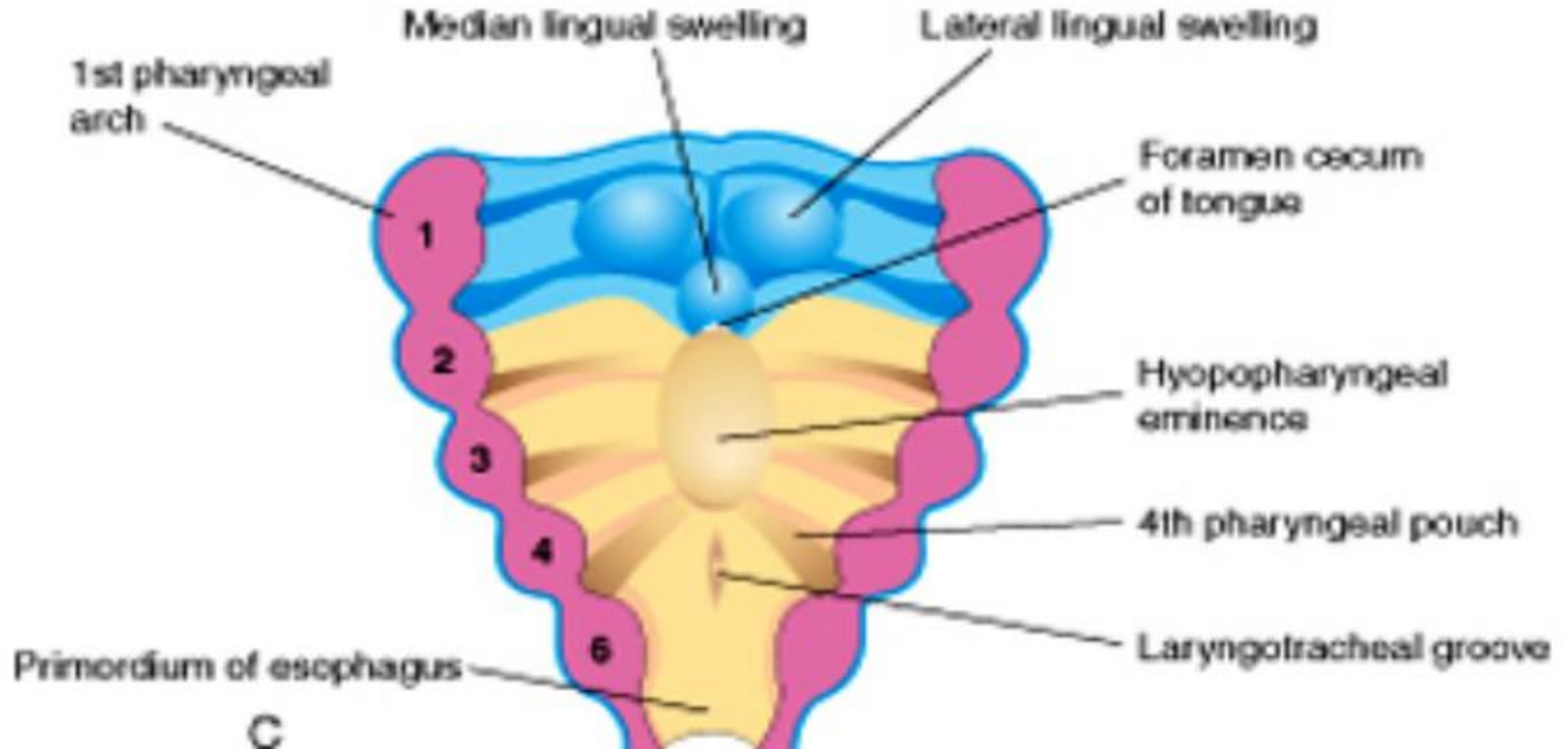
- **Respiratory Primordium** indicated at approximately 28 days by a median outgrowth-the **laryngotracheal groove**-from the **caudal end** of the **ventral wall** of the primordial pharynx



Introduction to Development of Respiratory System,,

- This primordium of the **tracheobronchial tree** develops **caudal to the *fourth pair of pharyngeal pouches***
- The endoderm lining the **laryngotracheal groove** gives rise to the **pulmonary epithelium and glands of the larynx, trachea, and bronchi**
- By the end of the 4th week, the laryngotracheal groove has evaginated to form a pouch like **laryngotracheal diverticulum** which is located ventral to the caudal part of **the foregut**.

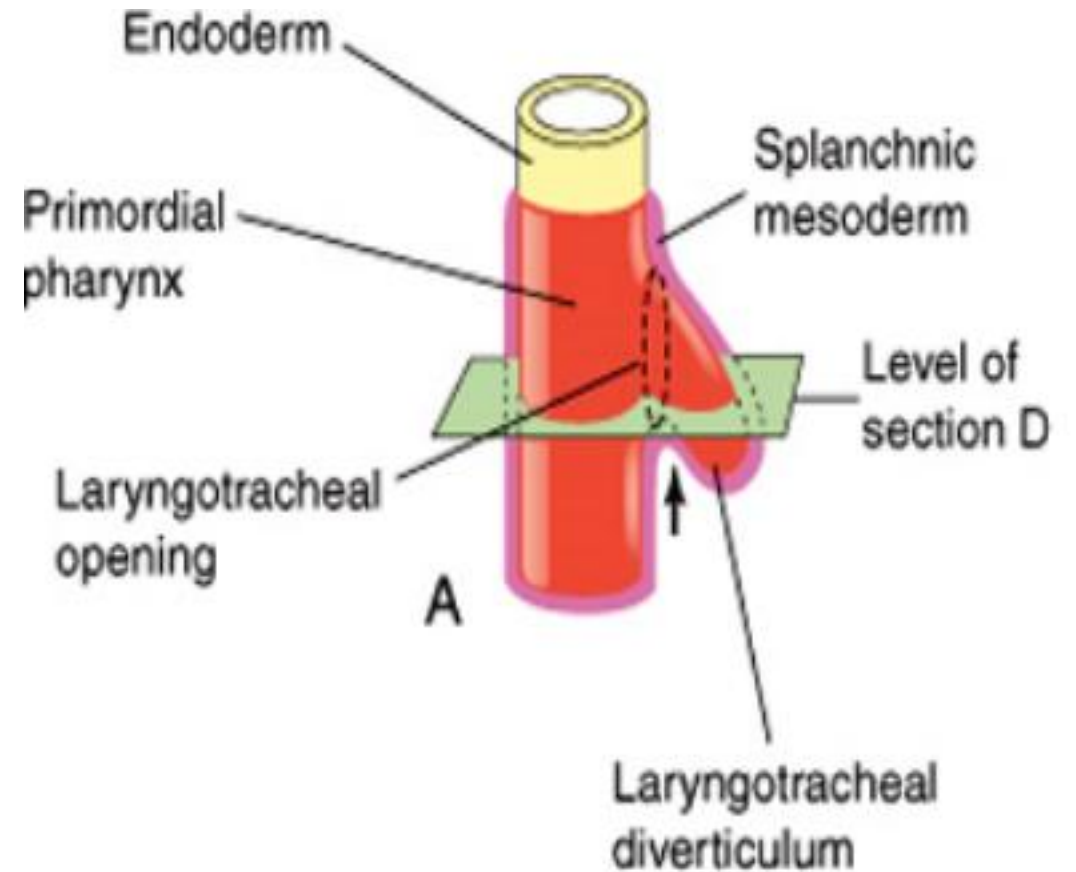
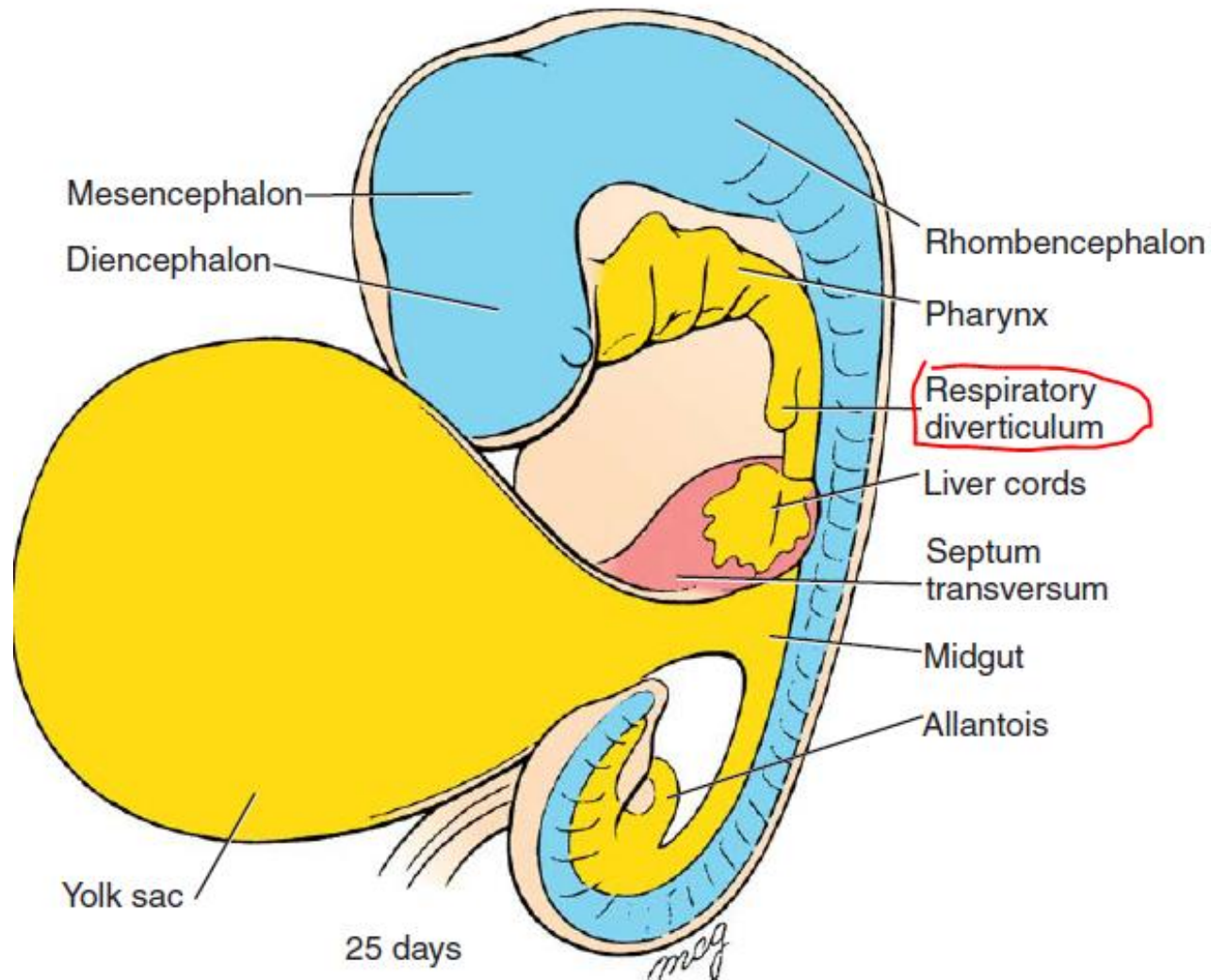
Introduction to Development of Respiratory System,,,



Respiratory bud

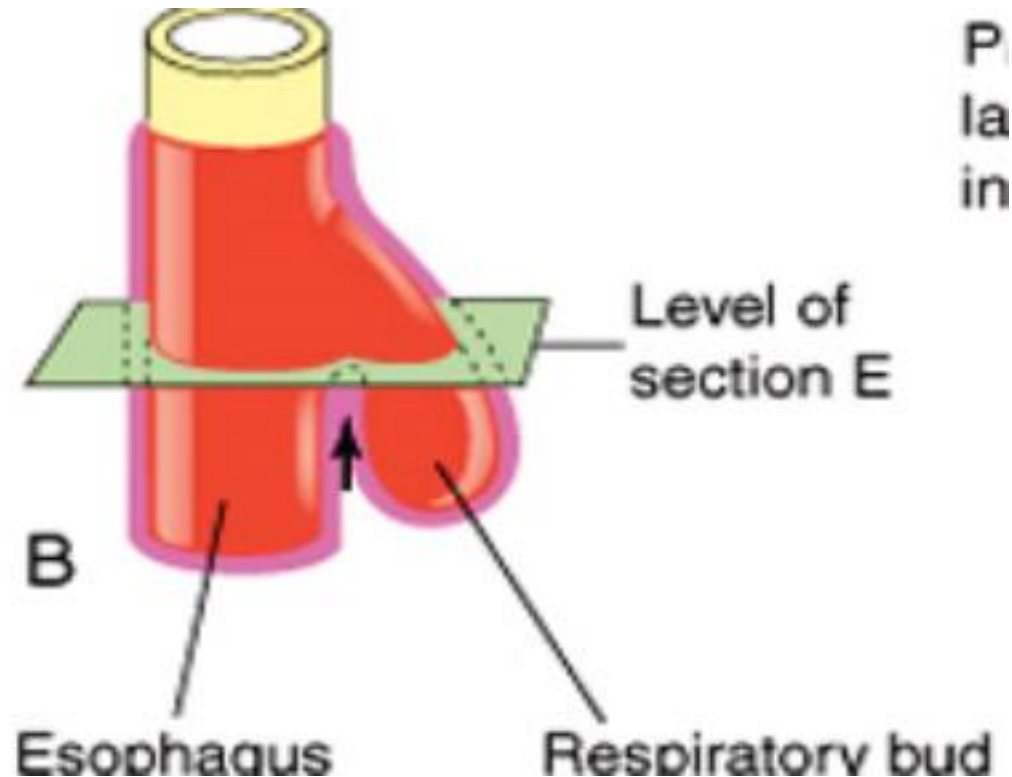
- The first sign of development is the formation of the respiratory (or laryngotracheal) diverticulum in the ventral wall of the primitive foregut during week 4.
- The distal end of the respiratory diverticulum enlarges to form the lung bud.
- The lung bud divides into two bronchial buds that branch into the main (primary), lobar (secondary), segmental (tertiary), and subsegmental bronchi.
- **The respiratory diverticulum** initially is in open communication with the foregut, but eventually they become separated by indentations of mesoderm, the tracheoesophageal folds.

Respiratory bud,,,



Laryngotracheal diverticulum

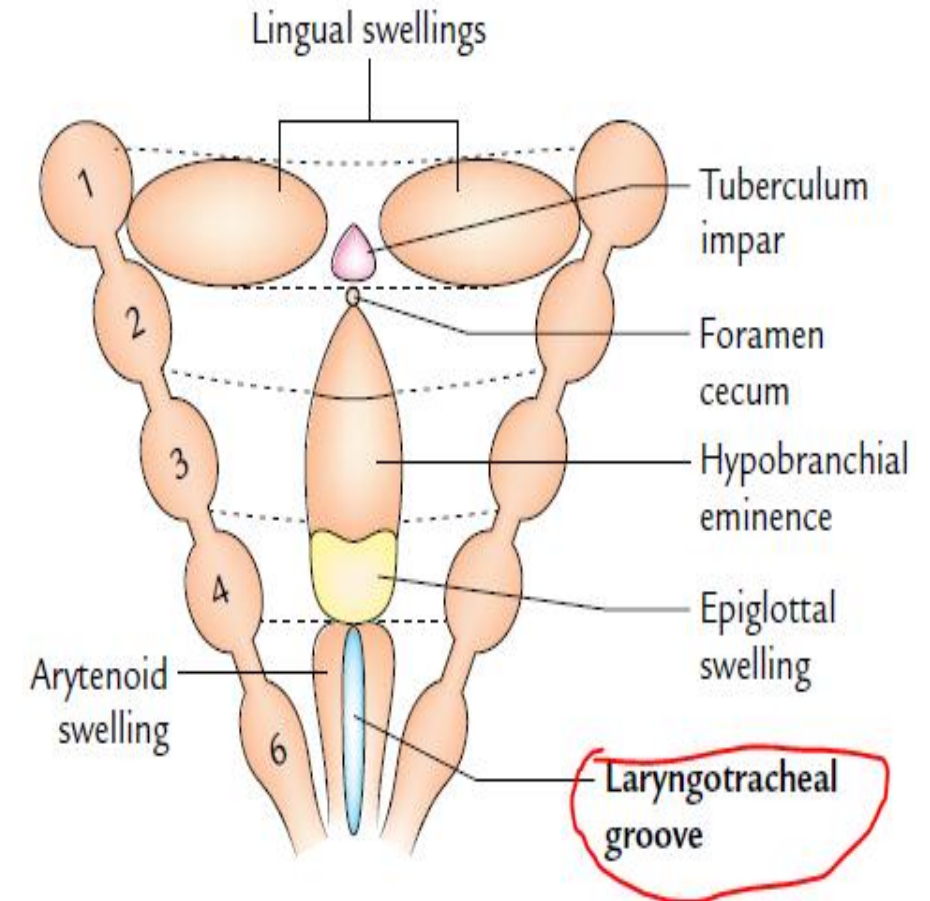
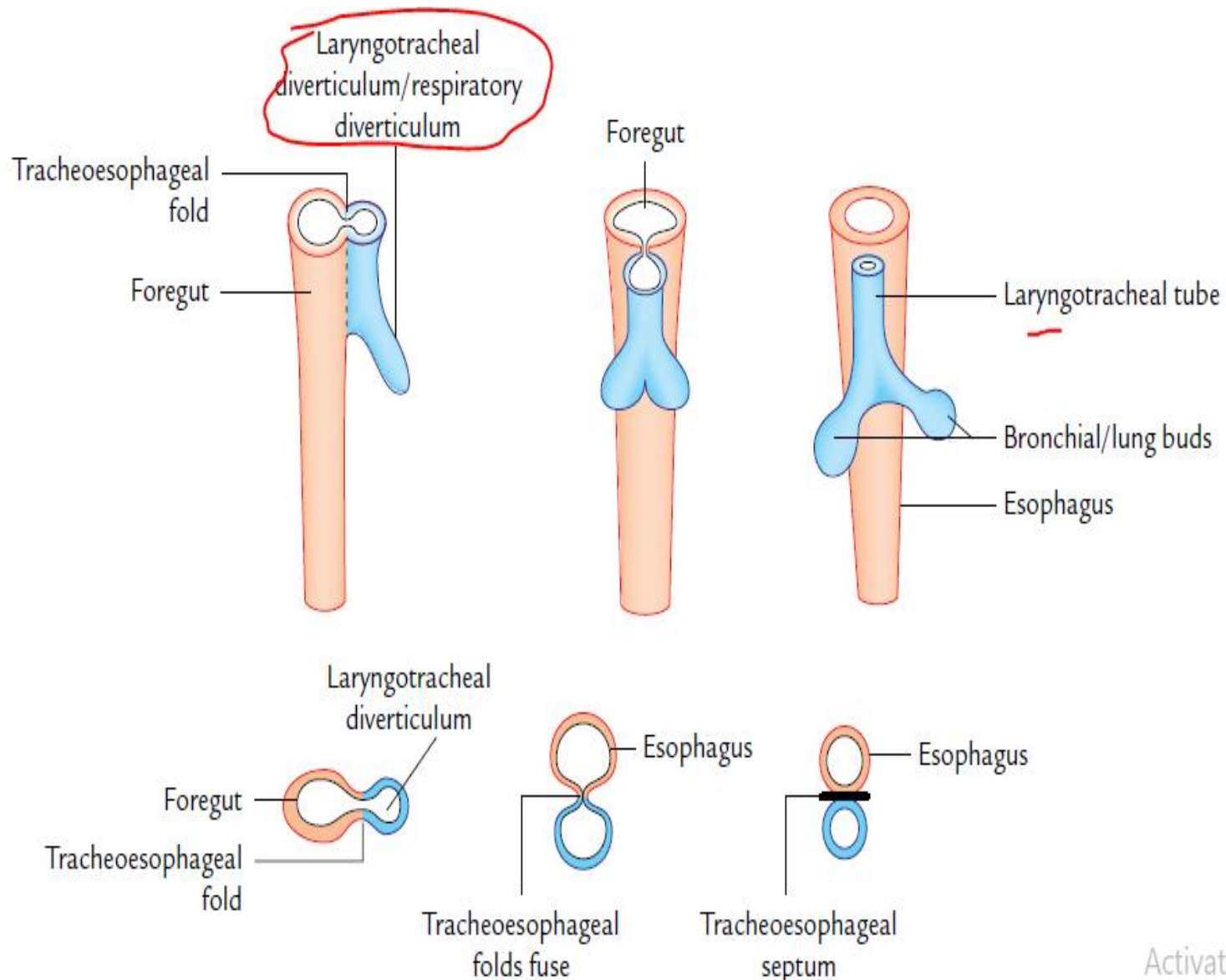
- **Respiratory bud:** laryngotracheal diverticulum **elongates**, it is invested with splanchnic mesenchyme and its **distal end enlarges** to form a globular **respiratory bud**.



Laryngotracheal diverticulum,,

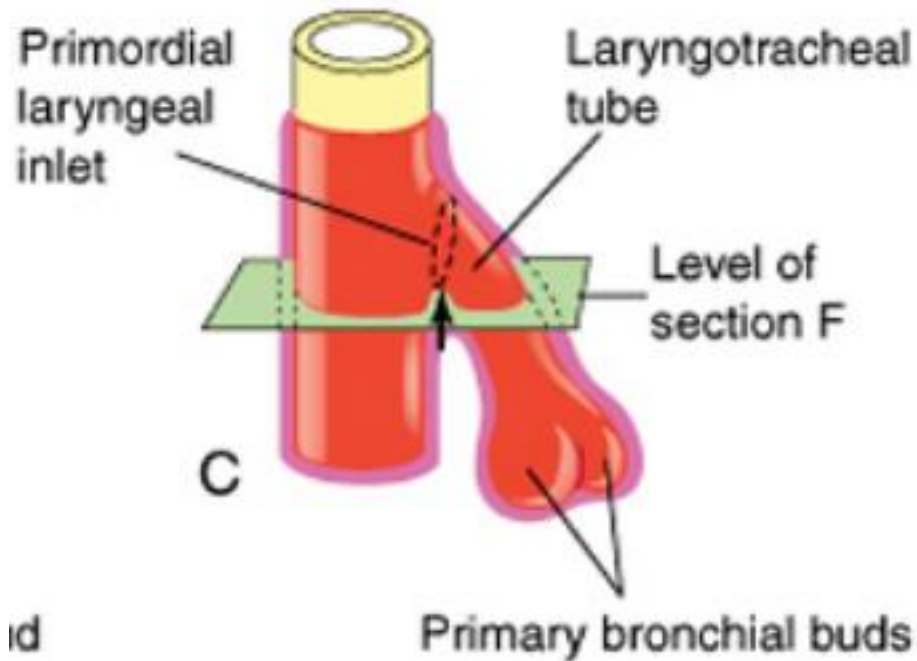
- The larynx develops from the cranial most part of **laryngotracheal diverticulum**.
- All the cartilages of the larynx (e.g., thyroid, cricoid, arytenoids, and cuneiform) **except epiglottis** develop from mesenchyme of fourth and sixth pharyngeal arch, .
- The epiglottis develops from the caudal part of **hypobranchial eminence**.
- The muscles of the larynx develop from the mesoderm of 4th & 6th pharyngeal arches.

Laryngotracheal diverticulum,,,



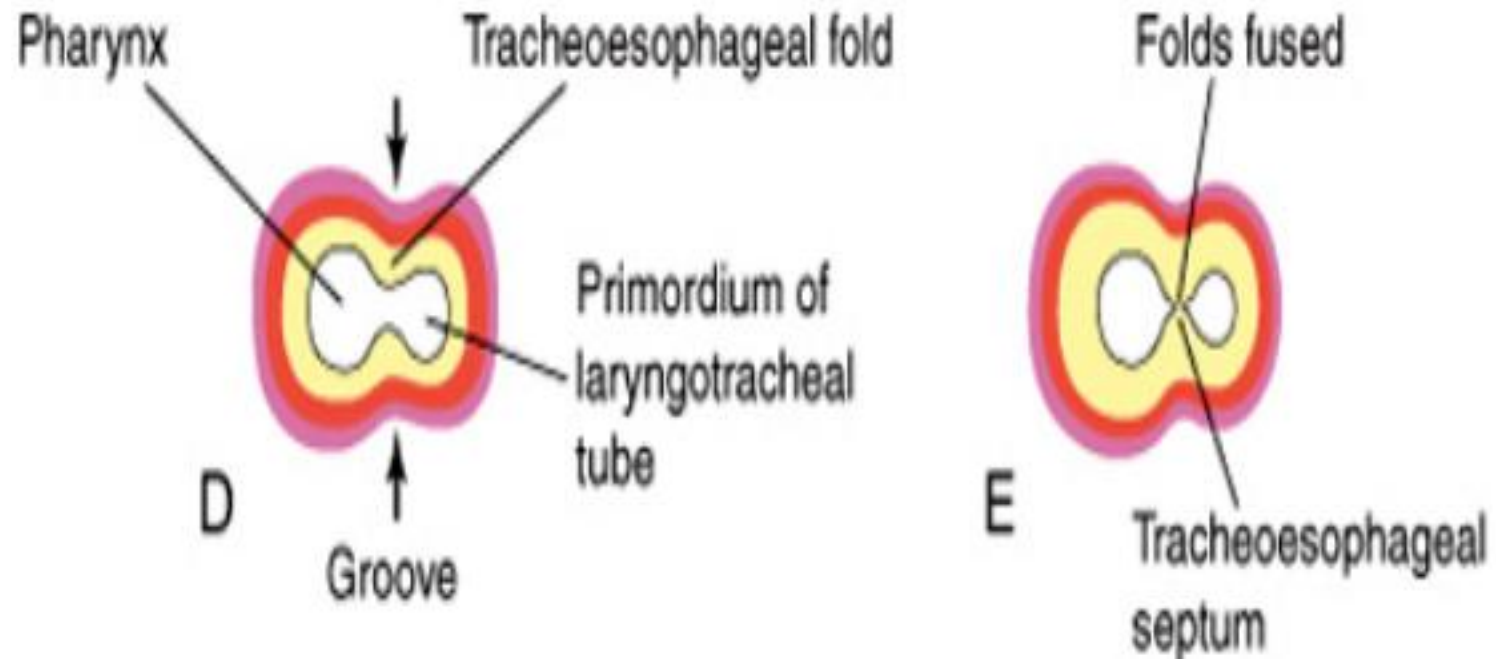
Laryngotracheal diverticulum,,,

- **Primordial laryngeal inlet:**
- The **laryngotracheal diverticulum** soon separates from the **primordial pharynx**; however, it maintains communication with it through the **primordial laryngeal inlet**.

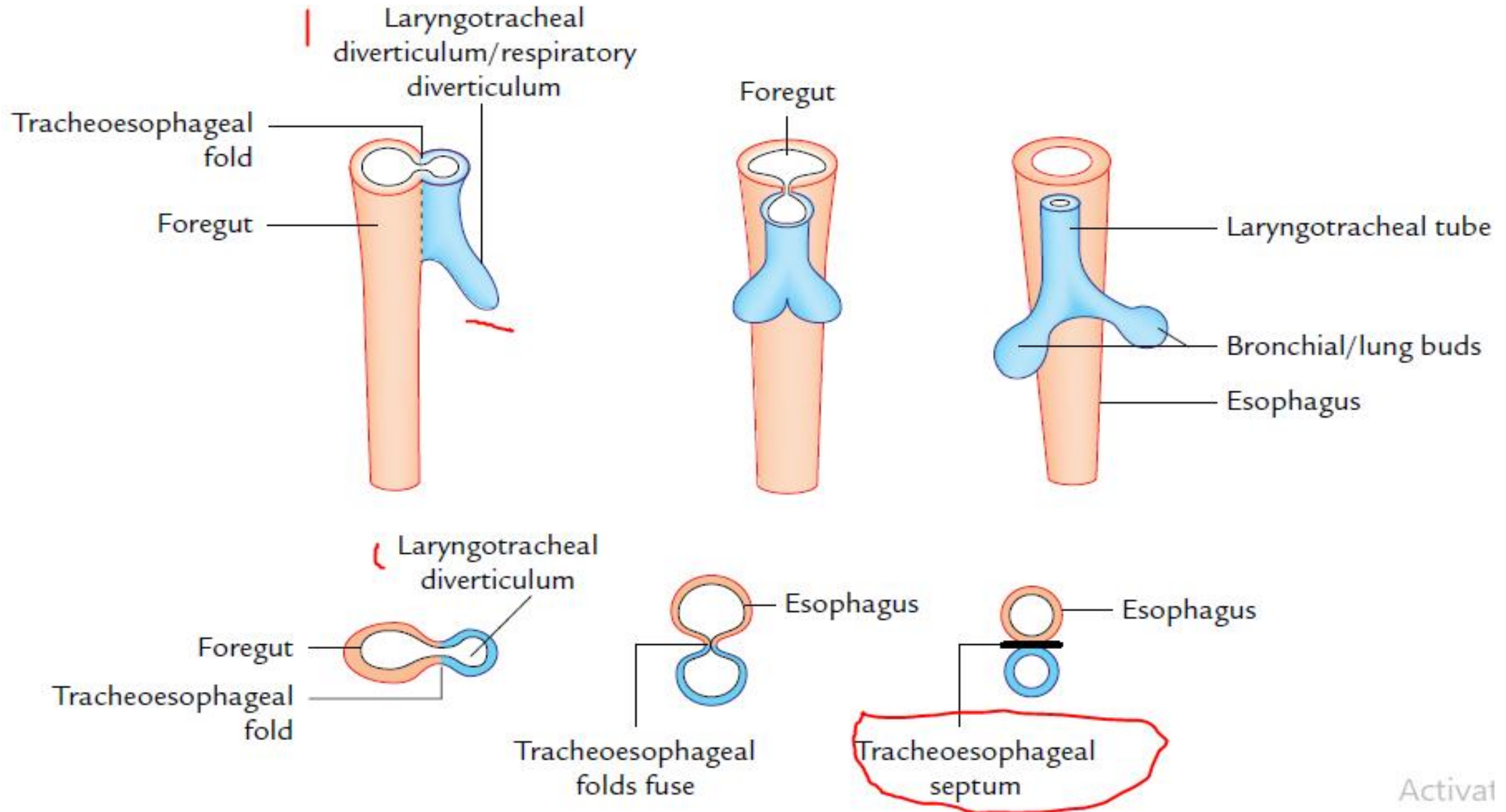


❖ Tracheoesophageal folds:

- Longitudinal **tracheoesophageal folds** develop in the laryngotracheal diverticulum.
- **Tracheoesophageal septum:** tracheoesophageal folds approach each other, and **fuse** to form a partition-the **tracheoesophageal septum**.

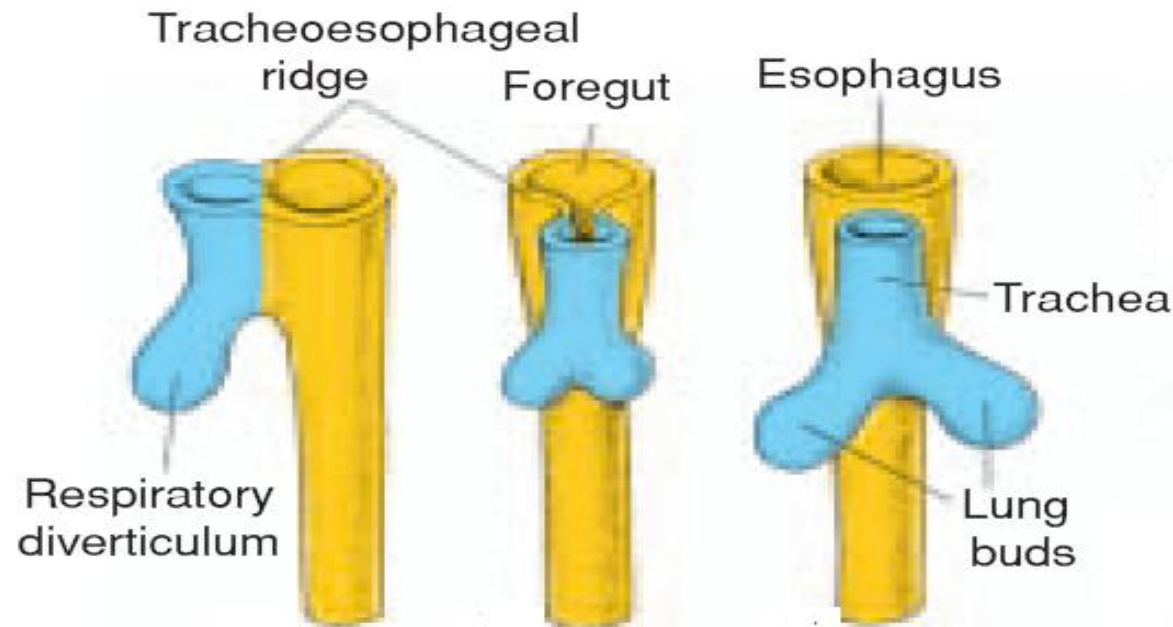


Tracheoesophageal folds:

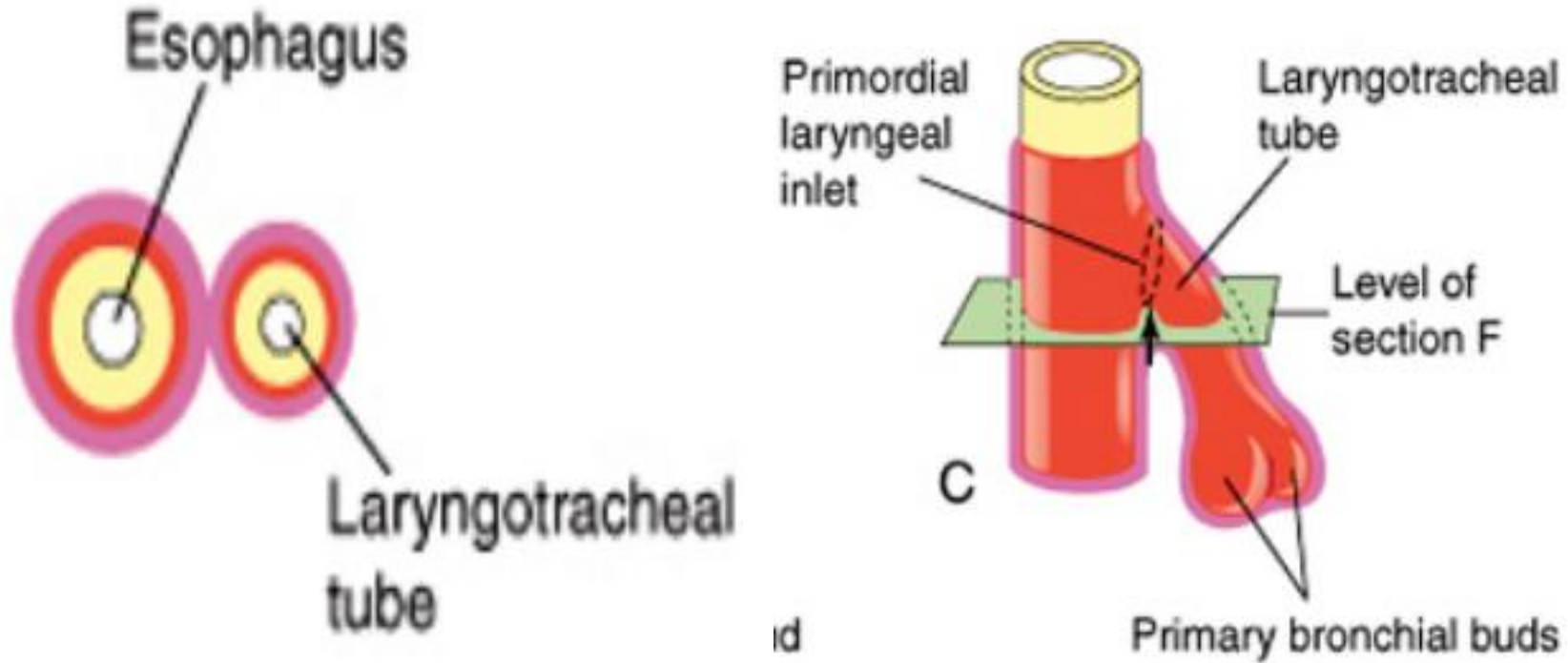


❖ Laryngotracheal tube:

- **Tracheoesophageal septum** divides the **cranial portion of the foregut** into
 - ventral part- **laryngotracheal tube** (*primordium of larynx, trachea, bronchi, and lungs*),
 - dorsal part -(primordium of oropharynx and esophagus)
- **primordial laryngeal inlet:** The opening of the laryngotracheal tube into the pharynx becomes the **primordial laryngeal inlet**



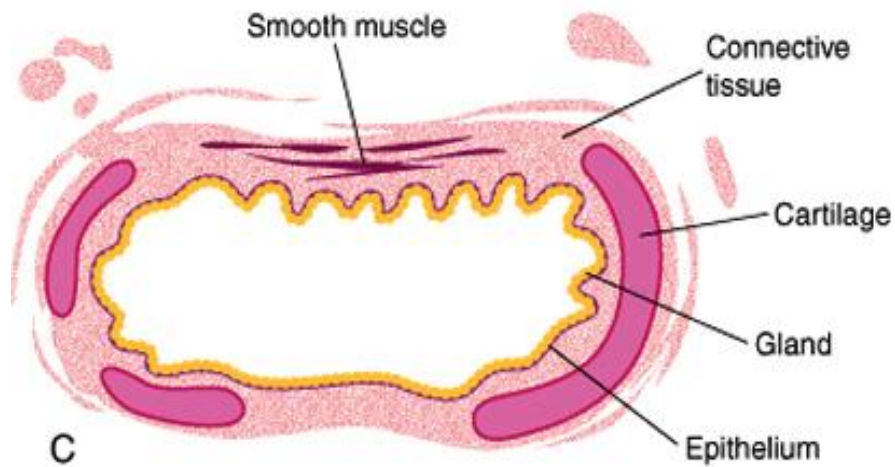
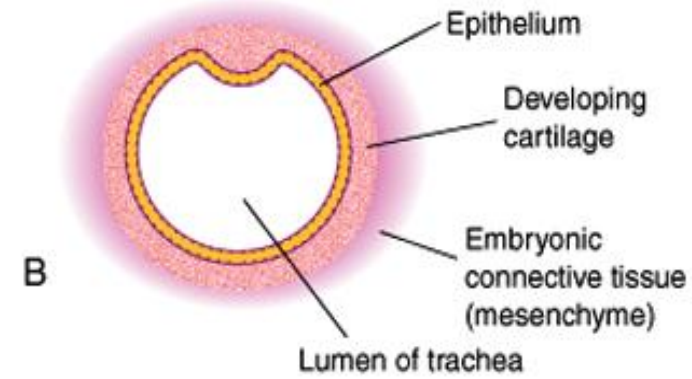
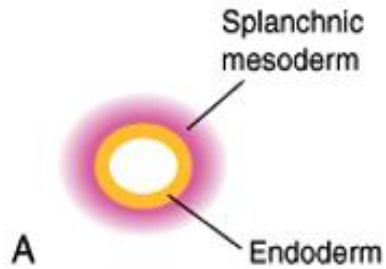
❖ Laryngotracheal tube:



Development of The Trachea

- **Epithelium** and **glands** of the trachea developed from **endodermal lining of the laryngotracheal tube** distal to the larynx .
- The endodermal lining of the laryngotracheal tube distal to the larynx differentiates into:
 - **the epithelium and glands of the trachea**
 - **the pulmonary epithelium.**
- **Cartilage, connective tissue, and muscles of the trachea:** derived from the **splanchnic mesenchyme** surrounding the laryngotracheal tube.

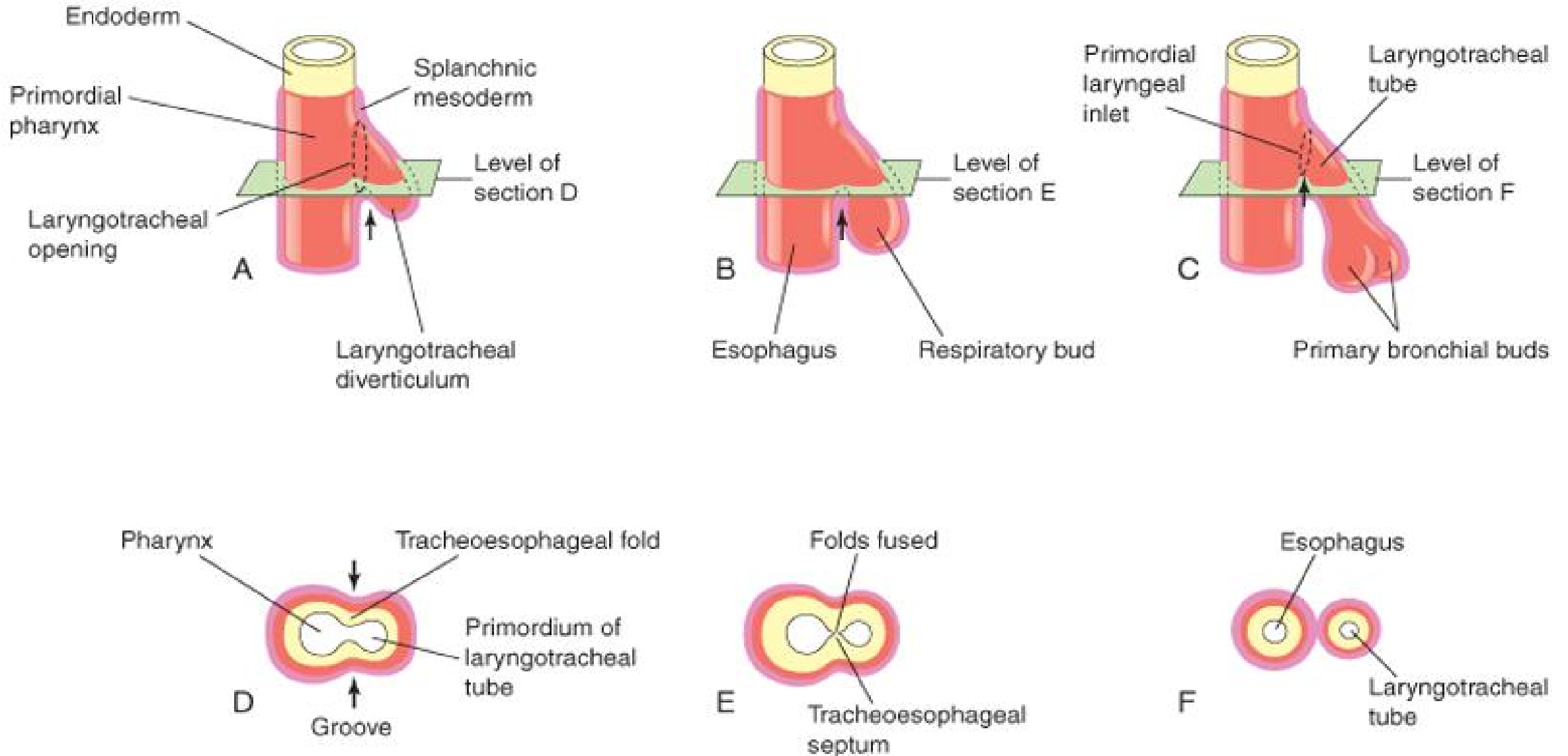
Development of The Trachea,,,



Development of the Bronchi and Lungs

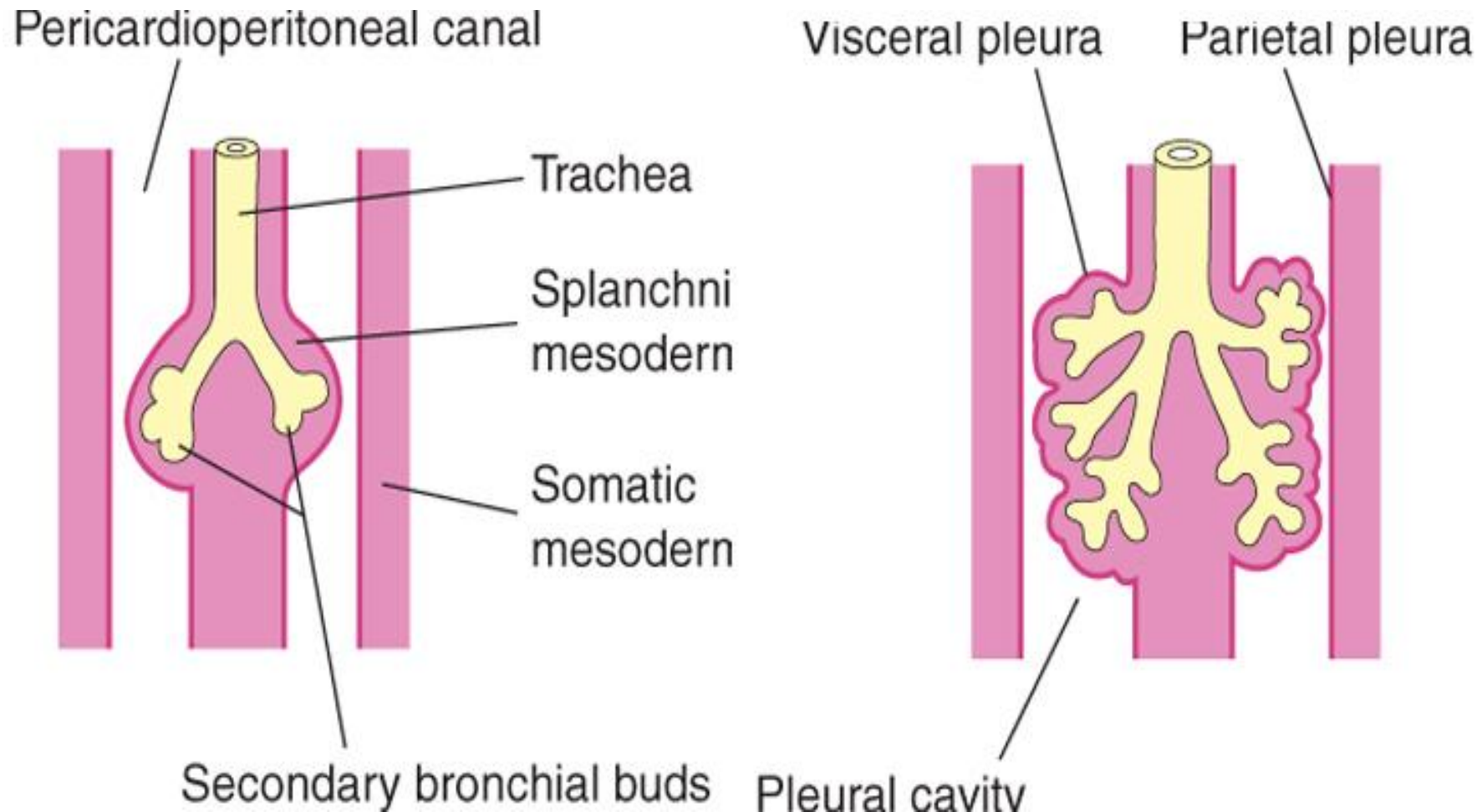
- **The respiratory bud (lung bud):** develops at the **caudal end** of the laryngotracheal diverticulum during **the 4th week** soon divides into two outpouchings-the **primary bronchial buds**
- These buds grow laterally into the **pericardioperitoneal canals** - the **primordia of the pleural cavities.**
- **Secondary and tertiary bronchial buds** soon develop.
- **Together with the surrounding splanchnic** mesenchyme, the bronchial buds differentiate **into the bronchi and their ramifications in the lungs.**

Development of the Bronchi and Lungs,,

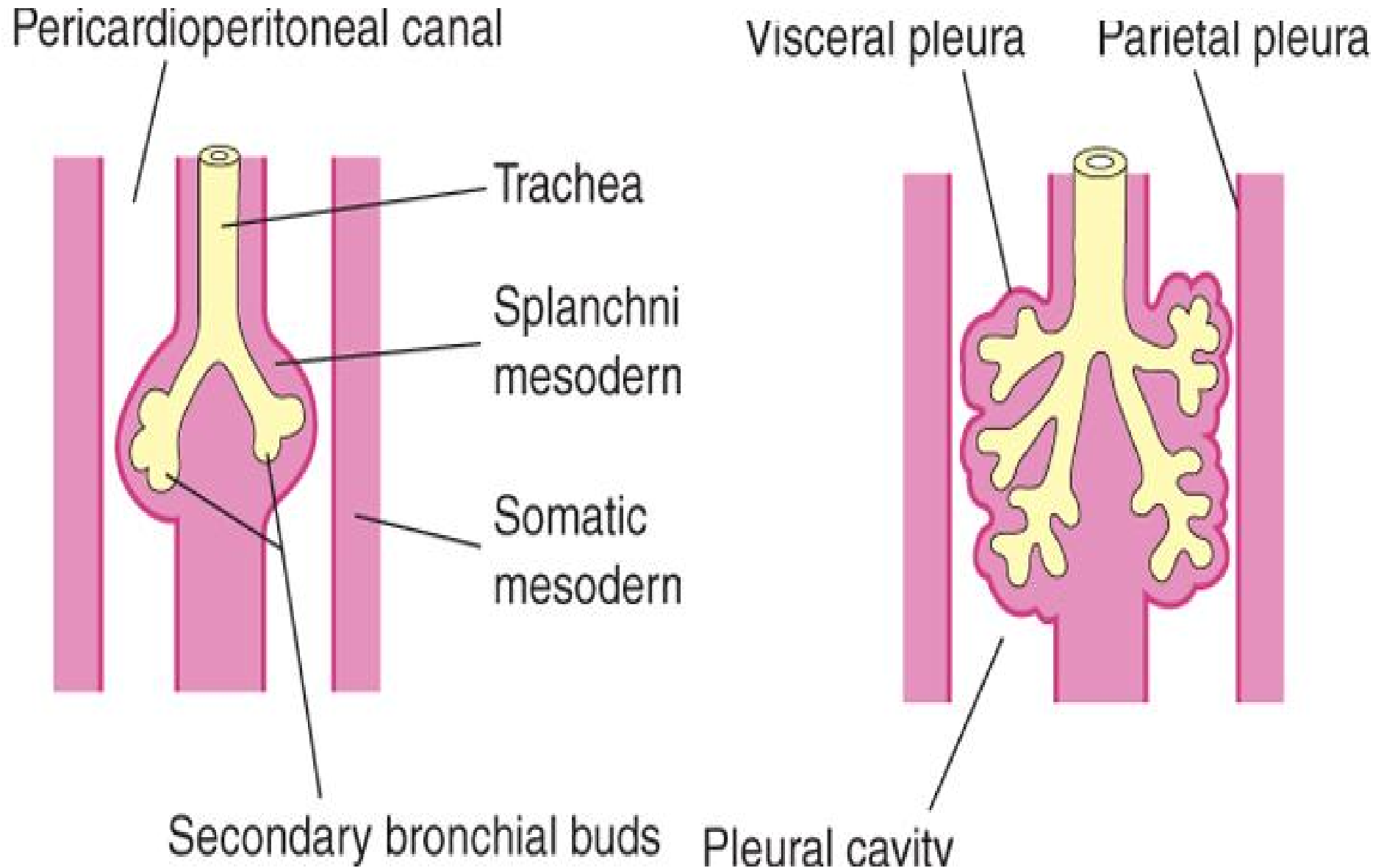


Development of the Bronchi and Lungs,,,

- Early in the **5th week**, the connection of each bronchial bud with the trachea enlarges to form the primordia of **main bronchi**



Development of the Bronchi and Lungs,,,



Development of the Bronchi and Lungs,,

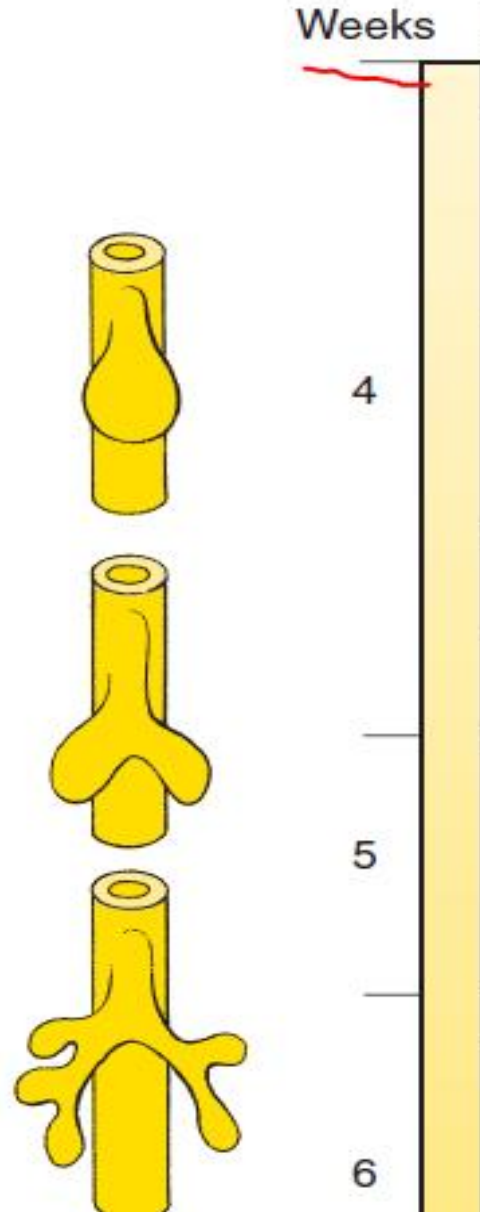
- The **segmental bronchi**, 10 in the right lung and 9 or 10 in the left lung, begin to form by the **seventh week**
- **Each segmental bronchus** with its surrounding mass of mesenchyme is the primordium of a **bronchopulmonary segment**.
- ***By 24 weeks***, approximately **17 orders of branches** have formed and **respiratory bronchioles** have developed
- An additional **seven orders** of airways **develop after birth**.
- **Cartilaginous plates** develop from the surrounding *splanchnic mesenchyme*.

Development of the Bronchi and Lungs,,,

Respiratory diverticulum forms

Respiratory diverticulum branches into left and right bronchial buds; stem of the diverticulum will differentiate into trachea and larynx

Branching yields secondary bronchial buds (three on right; two on left), which represent future lung lobes

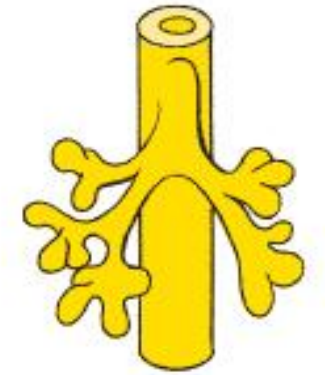


Branching yields tertiary bronchial buds (ten on right; eight on left), which represent the future bronchopulmonary segments

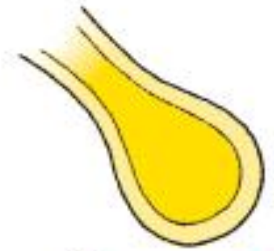
Terminal bronchioles form

Terminal bronchioles have formed respiratory bronchioles; the surrounding mesenchyme becomes highly vascular; first terminal sacs (primitive alveoli) form

Terminal sacs begin to differentiate into mature alveoli; alveoli continue to form through the 8th year



7



16



28



36

Birth

Developmental of Maturation of the Lungs

Maturation of the Lungs: is divided into four stages:

1. Pseudoglandular
2. Canalicular
3. Terminal sac
4. Alveolar

Developmental of Maturation of the Lungs,,

1. Pseudoglandular Stage (6 to 16 Weeks):

- The developing lungs somewhat histologically resemble **exocrine glands** during this stage
- **By 16 weeks**, all major elements of the lung have formed, **except those involved with gas exchange.**
- Respiration is **NOT** possible.

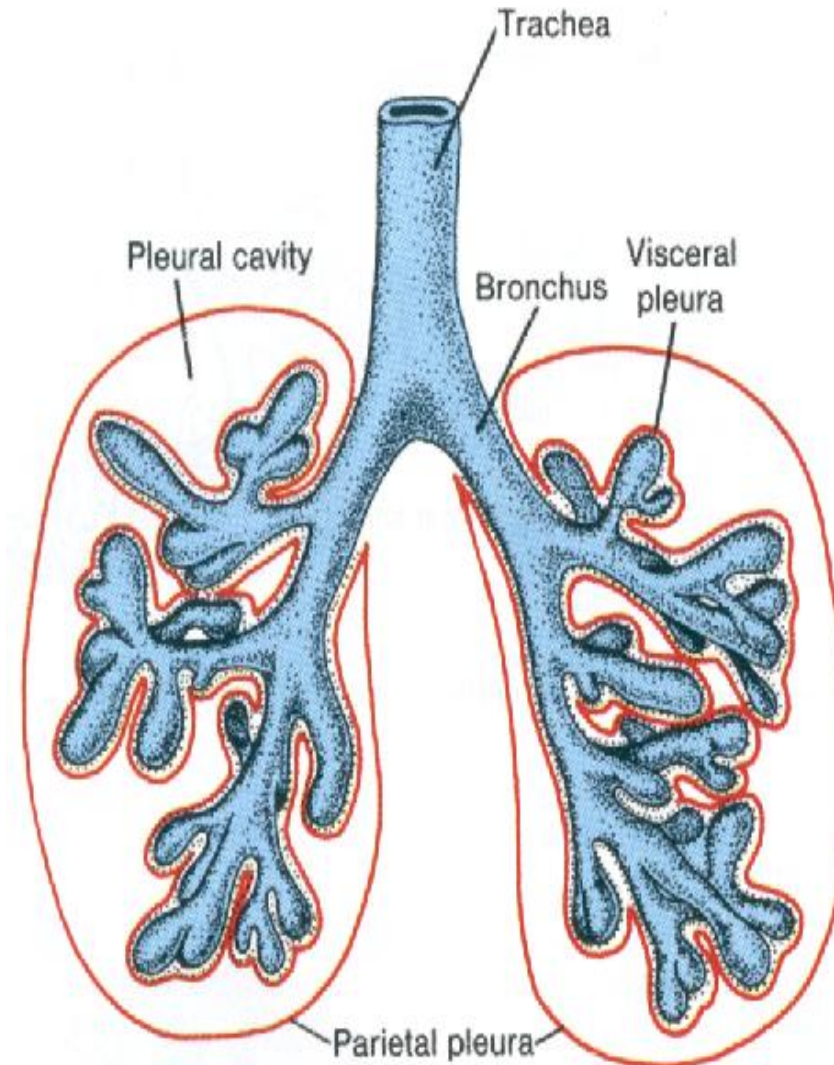
Developmental of Maturation of the Lungs,,

1. Pseudoglandular Stage (6 to 16 Weeks):

❑ In this stage:

- no respiratory bronchioles,
- no alveolar ducts,
- no alveoli,

❖ **Respiration is not possible;**
*hence, fetuses born during
this period are unable to
survive*



Developmental of Maturation of the Lungs,,

2. Canalicular Stage (16 to 26 Weeks)

- the lumina of the bronchi and terminal bronchioles become larger.
- the lung tissue becomes *highly vascular*
- **By 24 weeks**, each **terminal bronchiole** has given rise to two or more **respiratory bronchioles**
- each of **respiratory bronchioles** divides into **three to six** passages-**primordial alveolar ducts**.

Developmental of Maturation of the Lungs,,

- **2. Canalicular Stage (16 to 26 Weeks),,,**

- Respiration is possible **at the end of the canalicular stage** because some thin-walled **terminal sacs** (primordial alveoli) have developed at the ends of the respiratory bronchioles, and *the lung tissue is well vascularized.*
- Although a fetus born toward the end of this period may survive if given intensive care, it often dies because its respiratory and other systems are still relatively immature.

Developmental of Maturation of the Lungs,,

3. Terminal Sac Stage (26 Weeks to Birth):

- many more **terminal sacs or saccules** develop
- their *epithelium becomes very thin*.
- Capillaries begin to **bulge into these sacs** (developing alveoli).
- The intimate contact between epithelial and endothelial cells establishes the **blood-air barrier**, which permits adequate gas exchange for survival of the **fetus** if it is **born prematurely**.
- By 26 weeks, the terminal sacs are lined mainly by squamous epithelial cells of endodermal **origin-type I pneumocytes**-across which gas exchange occurs.

Development of the Bronchi and Lungs,,

3. Terminal Sac Stage (26 Weeks to Birth):,,

- Scattered among the squamous epithelial cells are rounded secretory epithelial cells-**type II pneumocytes**, *which secrete **pulmonary surfactant***
- **Surfactant** forms as a **monomolecular film** over the internal walls of the **alveolar sacs** and counteracts surface tension forces at the air-alveolar interface.
- This facilitates expansion of the saccules (primordial alveoli) by preventing atelectasis (collapse of saccules during exhalation)

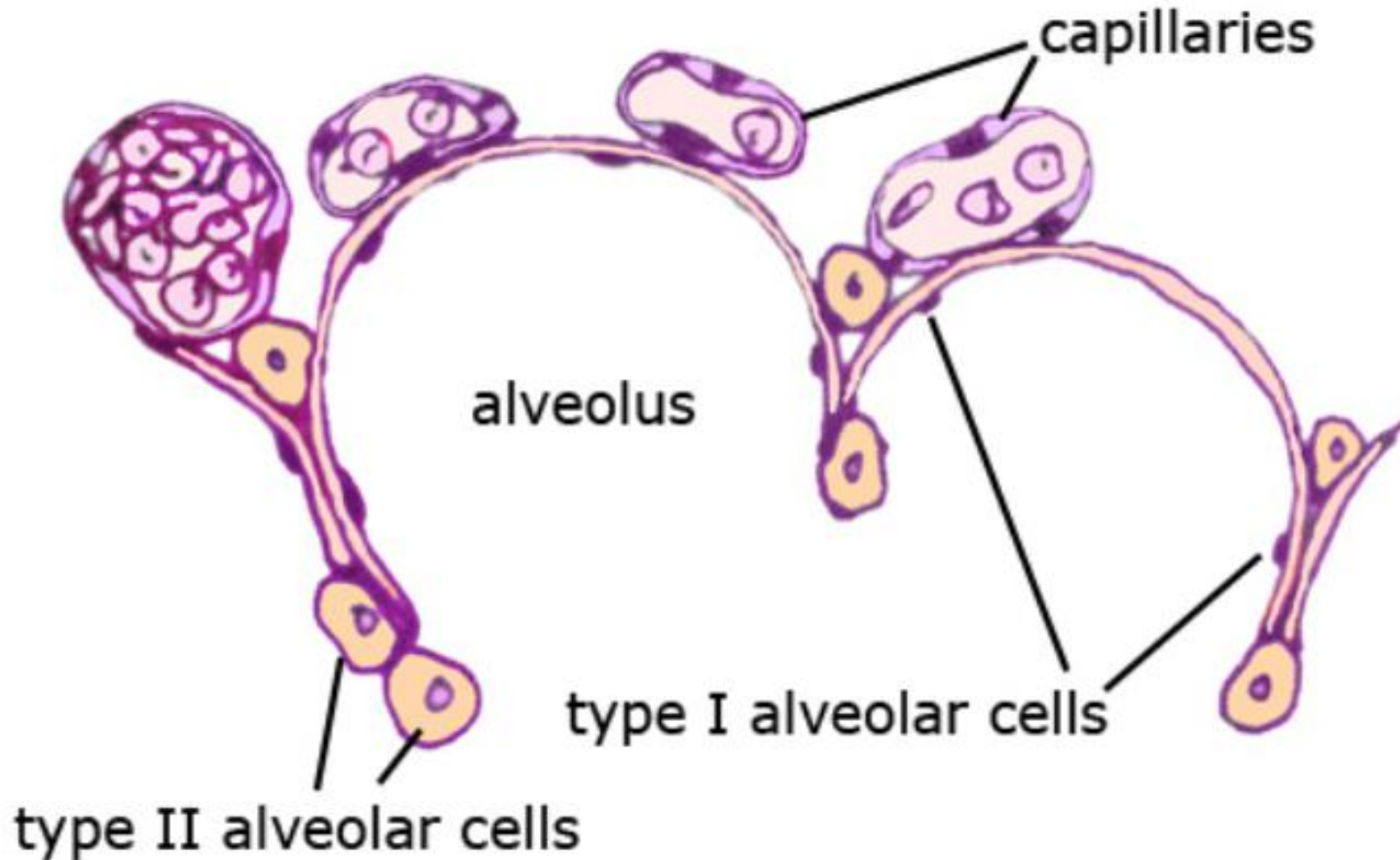
Developmental of Maturation of the Lungs,,

• 3. Terminal Sac Stage (26 Weeks to Birth):,,

• Type II Alveolar Epithelial Cells

- At the **end of the sixth month**, type II alveolar epithelial cells, **develop** and produce **surfactant**, a phospholipid-rich fluid capable of lowering surface tension at the air-alveolar interface.
- **Before birth the lungs are full of fluid** that contains a **high chloride concentration**, little **protein**, some **mucus** from the bronchial glands, and **surfactant** from the alveolar epithelial cells (type II).
- The **amount of surfactant** in the fluid **increases, particularly during the last 2 weeks** before birth.

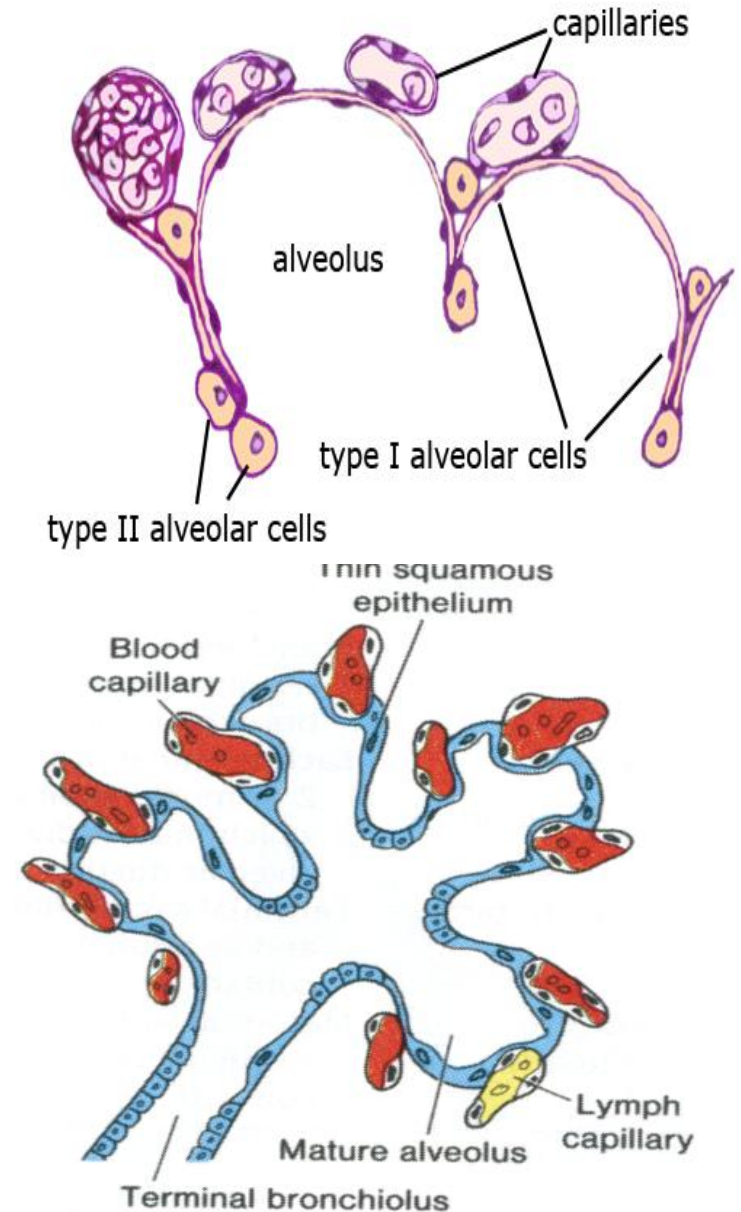
Developmental of Maturation of the Lungs,,,



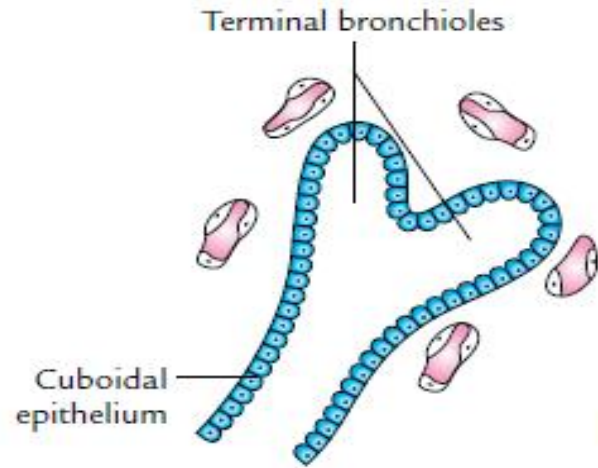
Developmental of Maturation of the Lungs,,,

4. Alveolar stage (period from 8th month of fetal life – 8 years after birth):

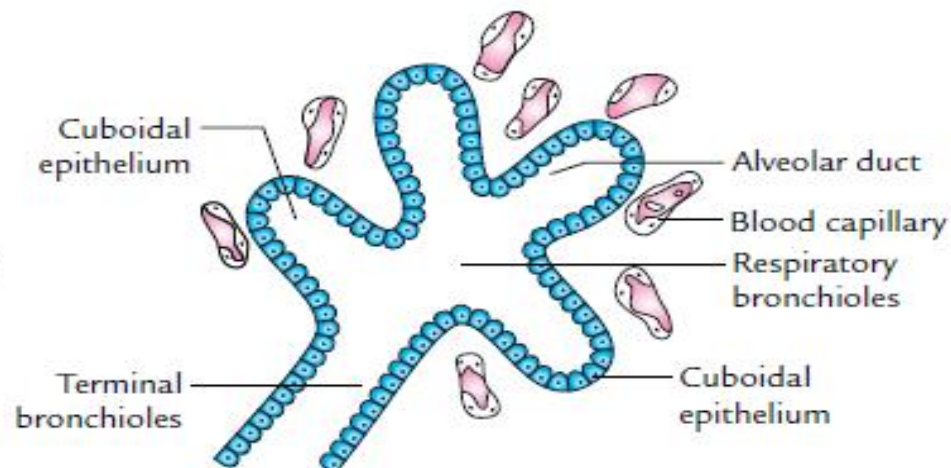
- ✓ **Mature alveoli** are formed, well developed alveolar endothelial (capillary) contact
- ✓ Thin squamous epithelial cells (alveolar epithelial cells type I)
- ✓ Cuboidal epithelial cells (alveolar epithelial cells type II)
- ✓ The surrounding capillaries protrude into mature alveoli



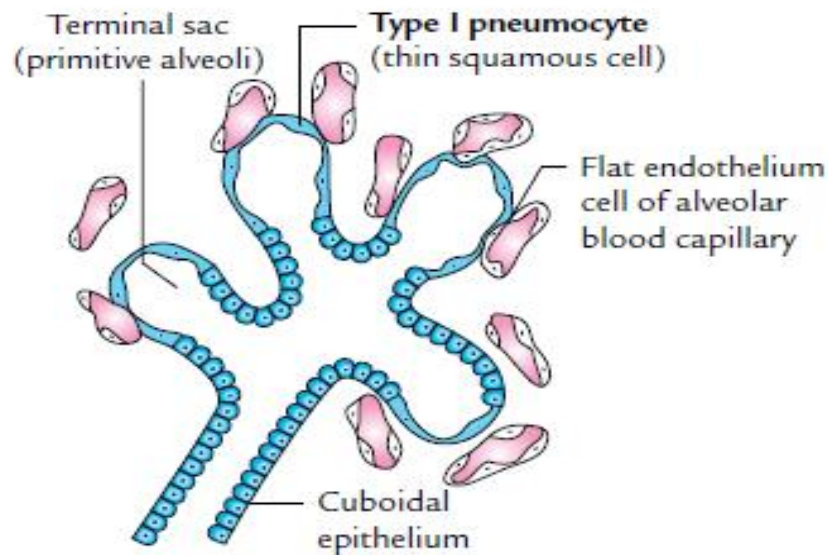
Developmental of Maturation of the Lungs,,



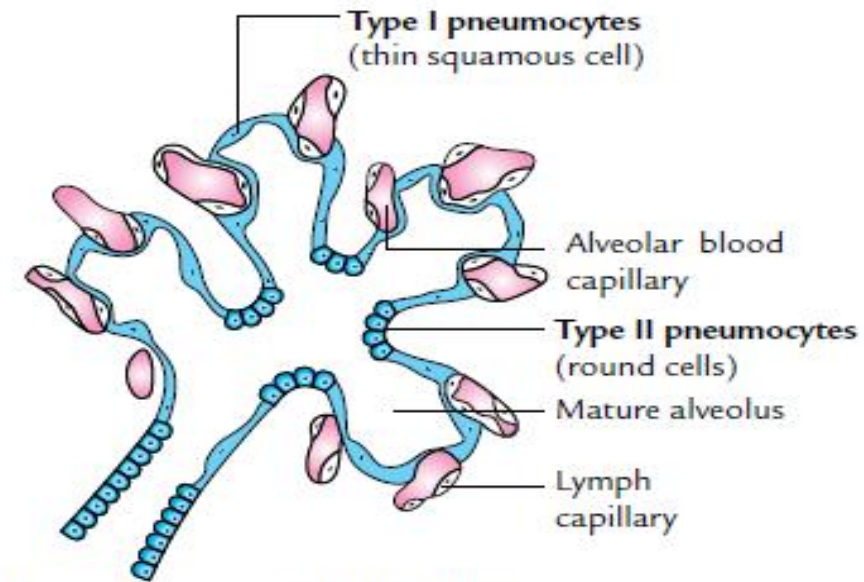
A Pseudoglandular stage



B Canalicular stage



C Saccular (terminal sac) stage



D Alveolar stage

Developmental of Maturation of the Lungs,,

Stages of maturation of lungs

Stage	Period	Development changes
Pseudoglandular stage	5–16 weeks	• Bronchial tree is formed up to terminal bronchioles • Elements of the bronchial tree involved in respiration (e.g., respiratory bronchioles, alveolar ducts, and alveoli) are not formed • Respiration is not possible at this stage; hence premature fetuses cannot survive if born at this stage
Canalicular stage	16–26 weeks	• Respiratory bronchioles and alveolar ducts are formed • Few alveolar sacs are also formed • Lung tissue is well vascularized • Fetus born at the end of the stage can survive if given intensive care
Terminal sac (saccular) stage	26 weeks to birth	• Large number of terminal sacs (primitive alveoli) develop • Capillaries bulge into developing sacs • Intimate contact develops between epithelium of sac and endothelium of capillary to permit adequate exchange of gases for survival of fetus, if born prematurely. Fetuses born at this stage survive
Alveolar stage	8 months to 8 years	• Formation of definitive (true) alveoli with an increase in their number • Type II pneumocytes produce a sufficient amount of surfactant • Free exchange of gases across the blood–air barriers formed by epithelium of alveoli and endothelium of capillaries

Developmental of Maturation of the Lungs,,

- **N.B.** The transition from dependence on the placenta for gas exchange to autonomous gas exchange requires the following adaptive changes in the lungs:
 - 1 .Production of surfactant in the alveolar sacs
 - 2 .Transformation of the lungs from secretory into gas exchanging organs
 - 3 .Establishment of parallel pulmonary and systemic circulations

Developmental of Maturation of the Lungs,,

Growth Of The Lung After Birth Is Due To:

- 1- increase in the **number** of *respiratory bronchioles*
- 2- increase in the *number* of **alveoli** (new alveoli are formed in the first 8 years of postnatal life)

NB -The increase in the size of the lung after birth is not due to increase in the size of the alveoli

- The pleura (parietal & visceral) develops from the **mesoderm** of lateral plate **mesoderm**

Congenital anomalies of respiratory system

Anomalies of trachea

❖ **Tracheoesophageal Fistula(TEF)** : A fistula (abnormal passage)

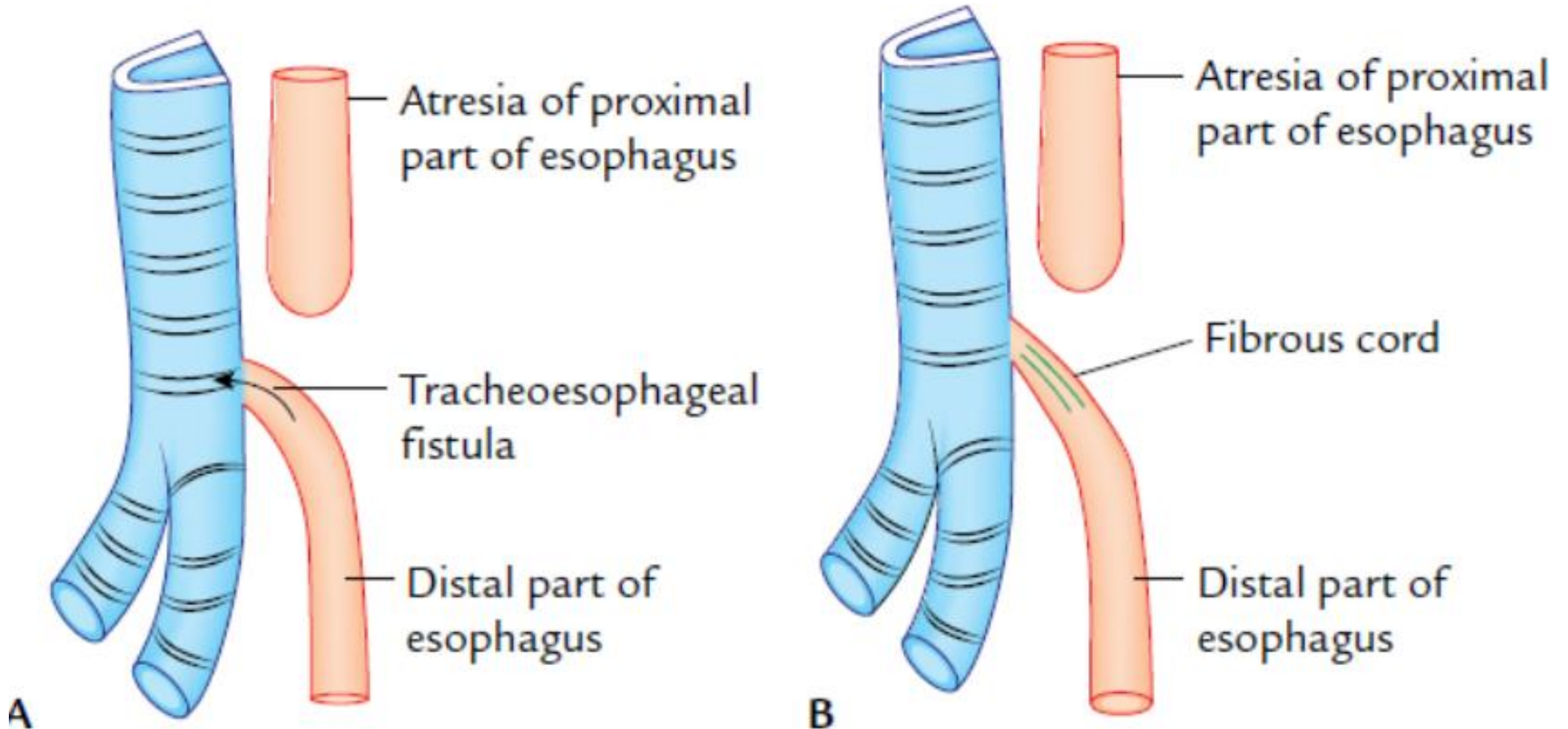
between the trachea and esophagus ,

- occurs once in 3000 to 4500 live births, most **affected infants are males**,
- In more than 85% of cases, the tracheoesophageal fistula (TEF) is associated with esophageal atresia.
- TEF results from incomplete division of the cranial part of the foregut into respiratory and esophageal parts during the fourth week.

Anomalies of trachea,,,,

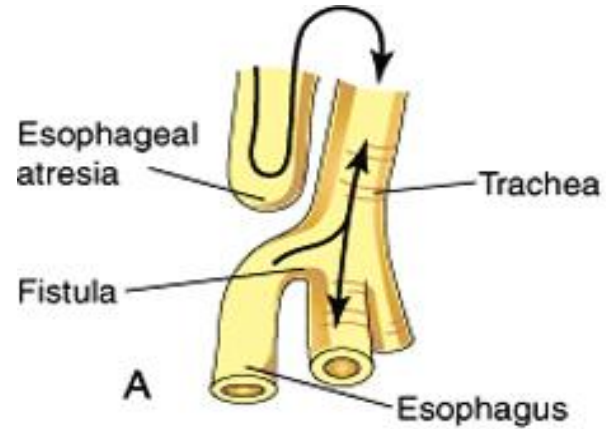
- *TEF is the most common anomaly of the **lower respiratory tract**.*
- The following are the forms of TEF
 1. Upper part of the esophagus ends in a blind pouch and lower part communicates with the trachea (85–90%)
 2. As type (a) but the tracheoesophageal communication/canal is replaced by a fibrous cord

Anomalies of trachea,,,,



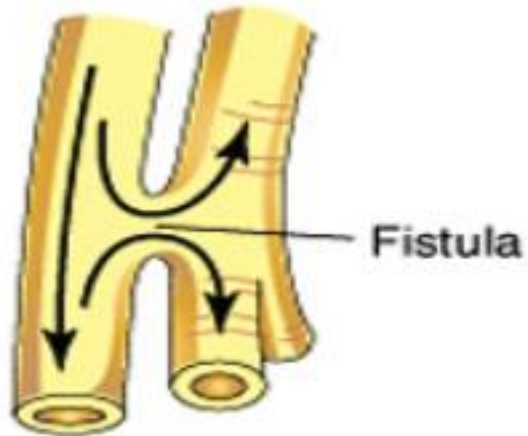
Anomalies of trachea,,,,

Tracheoesophageal Fistula



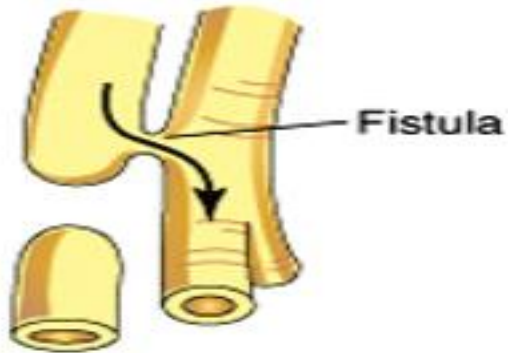
A is associated with TEF in more than 85% of cases

Possible directions of the flow of the contents are indicated by *arrows*



Fistula between the trachea and esophagus

Anomalies of trachea,,,,



Air cannot enter the distal esophagus and stomach



Air can enter the distal esophagus and stomach,

Respiratory distress syndrome

- Respiratory distress syndrome (RDS): This clinical condition affects about 2% of live newborn infants.
- As the name suggests infants born with this condition develop rapid, labored breathing shortly after birth.
- This condition is common in premature infants.
- The most common cause of RDS is deficiency of surfactant.

Respiratory distress syndrome,,,

❖ Respiratory Distress Syndrome (Hyaline Membrane Disease)

- It occurs in the *premature infants*, due to *deficient amount of the surfactant*, leads to partial *collapse of alveoli*
- The partially collapsed alveoli contain fluid with high protein content → forming **lamellar bodies** & many **hyaline membranes** so called → **Hyaline membrane disease**



**Thank you
for your attention**